

Dataset Analysis Report – Titanic Dataset

1. Dataset Overview

The Titanic dataset contains information about passengers aboard the Titanic. It consists of 891 records and 12 features. The primary goal of the dataset is to predict passenger survival based on various attributes.

2. Feature Classification

- **Numerical Features:** Age, Fare, SibSp, Parch
- **Categorical Features:** Sex, Embarked, Ticket
- **Ordinal Features:** Pclass (1st, 2nd, 3rd class)
- **Binary Features:** Survived (0 = No, 1 = Yes)

3. Target Variable

The target variable is Survived, which indicates whether a passenger survived or not. This makes the dataset suitable for a classification problem.

4. Data Quality Observations

- Missing values present in Age, Cabin, and Embarked
- Cabin has a high percentage of null values
- Class imbalance exists in the Survived column
- Categorical variables require encoding before modelling

5. Machine Learning Readiness

The dataset is suitable for machine learning after preprocessing steps such as:

- Handling missing values
- Encoding categorical variables
- Feature scaling (if required)

Conclusion

Understanding the dataset structure and data types helps in selecting appropriate preprocessing techniques and improves model performance